

Ibrahim Weziza

ML Engineer

[LinkedIn](#) | [Phone: +20 1552997580](tel:+201552997580) | [Email: i.weziza@gmail.com](mailto:i.weziza@gmail.com) | [GitHub](#)

[Location](#): Zahraa Nasr City (ready to reallocate) | [Military Status](#): **Exemption**

Education

- Bachelor of Engineering (BE), Electronics and Communication, Mansoura University (2019 – 2024)
- Graduation project: **Diabetic Retinopathy Classification**
Developed a deep learning model for diabetic retinopathy detection, achieving 85% classification accuracy on the APTOS 2019 dataset.
Applied image preprocessing techniques to enhance model generalization on 3,500+ retinal images.
Utilized transfer learning with EfficientNet-B3 for improved performance in multi-class disease staging.
Deployed the trained model as a streamlit-based web application for automated, real-time diagnosis support.

Projects

AI Reporter Agent

- Developed an AI multi-agent system using CrewAI to automate product research and analysis workflows.
- Created specialized agents for query generation, commercial web search via Tavily API
- Integrated LLMs for information extraction, ensuring relevant and high-confidence product data retrieval.
- Automated the generation of structured HTML reports summarizing product specifications, pricing, and availability for insights delivery.
- AI Reporter Agent | Tools: CrewAI, Tavily API, LangChain, LLMs

Fine-Tuning DeepSeek R1 on Arabic Data

- Fine-tuned DeepSeek R1 (7B) LLaMA-based model using LoRA adapters, achieving a 92% accuracy on Arabic text extraction tasks. Prepared custom Pydantic schemas for structured JSON extraction from Arabic stories.
- Implemented a custom generation pipeline for entity and metadata extraction from Arabic texts.

Side Data Science Projects

- Built multiple AI and data science projects leveraging Hugging Face Transformers for zero-shot learning, finetuning, and text classification applications.
- Applied LLMs for sentiment analysis, multi-class classification, prediction, and language understanding tasks.
- Delivered end-to-end solutions covering data collection, preprocessing, feature engineering, model training, hyperparameter tuning, and evaluation.
- Deployed models into production-ready pipelines and interactive apps, ensuring reliable performance and practical business insights.

Skills

Machine Learning, Deep Learning, Transfer Learning, AI Agents, Prompt Engineering, RAG
Data Cleaning, EDA, Feature Engineering, Arabic NLP, Model Deployment, API Integration
LangChain, Hugging Face, Transformers, Python, SQL, Power BI, Tableau, Git, GitHub

Certificates

- AI Engineer bootcamp - NLP, Transformers, LLMs, LangChain, APIs (2025)
- Machine Learning Specialization - *DeepLearning.AI* (2024)
- Mathematics for Machine Learning and Data Science Specialization - *DeepLearning.AI* (2024)
- Database and SQL for Data Science - *IBM* (2024)
- Share Data Through the Art of Visualization - *Google* (2023)
- Ask Questions to Make Data-Driven Decisions – *Google* (2023)
- Data Cleaning - *Kaggle* (2023)
- Python for Data Science - *University of Michigan* (2023)

Language

- Arabic
- English

Volunteering

Business coordinator at "Student for Egypt faculty of Engineering, Mansoura University"